# **Appendix 3** *for:* Beyond the average phenotype: individual variation alters mechanistic forecasts of climate responses

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# Additional Results

## 1 Jensen bias for males

**Figure S3.1** shows the spatial distribution of Jensen bias ($J$) between population-average and individual-level prediction surfaces for males under current European climate. While the population-average DEB parameterisation underestimated most performance metrics for females (**Fig. 2**), the pattern for males was largely opposite: the population-average parameterisation overestimated most metrics, including reproductive energy ($|J| = 2315$ J), basking time ($|J| = 286$ h) and foraging time ($|J| = 398$ h). In relative terms, male bias was largest for reproductive energy (normalised $|J| = 9.1\%$), $R_0$ (7.4%) and food intake (6.9%). Overestimation dominated for body size, reserve, food, reproductive energy and basking metrics ($J > 0$ in 57–96% of cells), whereas maximum and minimum body temperature showed predominantly negative bias ($J < 0$ in 84–93% of cells). The magnitude and direction of Jensen bias therefore depended strongly on both sex and ecological performance metric.

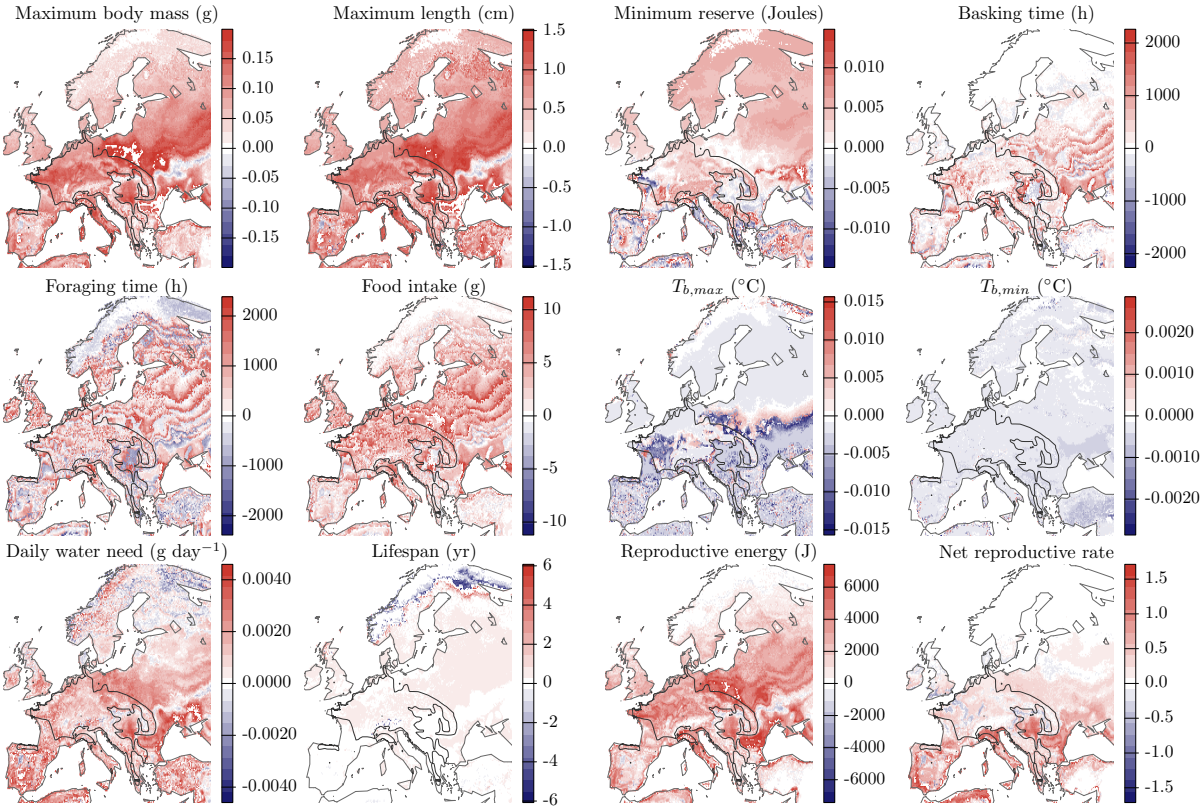



**Figure S3.1:** Jensen bias ($J$) between population-average and individual-level prediction surfaces for males. Values close to zero indicate close agreement; departures from zero indicate systematic differences in the spatial pattern. The colour scale is symmetric and centred at zero. The black outline delimits the current distribution of *I. alpestris* (IUCN 2021). $T_b$, body temperature.

# 2 Coefficient of variation maps

The coefficient of variation (CV) across the 37 individual-level predictions quantifies the degree of physiological heterogeneity independent of Jensen bias. High CV values identify regions and metrics where individual variation most strongly influences predicted outcomes.

## 2.1 Females

For females, CV was highest toward the range margins for most performance metrics (**Fig. S3.2**), indicating that the ecological consequences of individual physiological variation were spatially structured rather than uniform across the landscape.

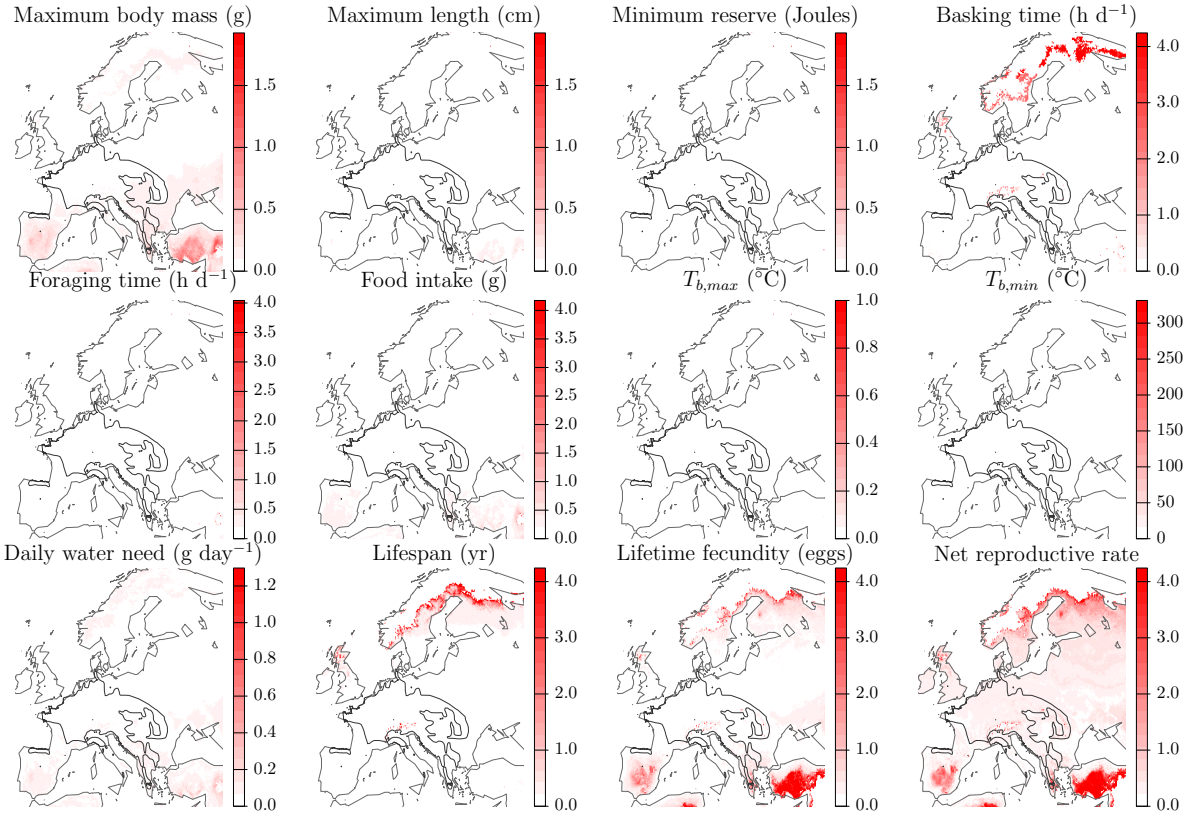



**Figure S3.2:** Coefficient of variation (CV) across individual-level predictions for females. Each panel shows the spatial distribution of CV for one performance metric. Higher values indicate greater individual-level variation in that metric across the geographic range. The black outline delimits the current distribution of *I. alpestris* (IUCN 2021). $T_b$, body temperature.

## 2.2 Males

A similar spatial pattern was observed for males (**Fig. S3.3**), with CV values peaking at range margins for most metrics, consistent with the pattern observed in females.

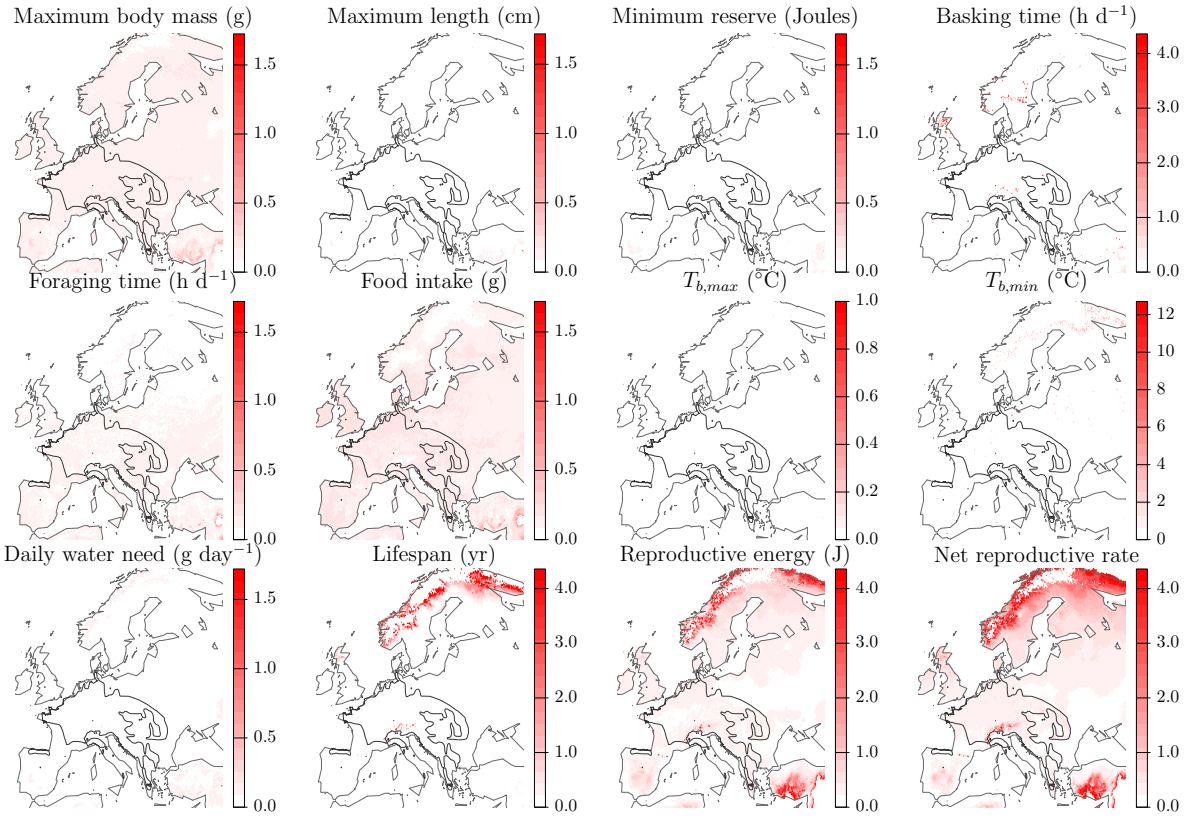



**Figure S3.3:** Coefficient of variation ($CV$) across individual-level predictions for males. Each panel shows the spatial distribution of $CV$ for one performance metric. Higher values indicate greater individual-level variation in that metric across the geographic range. The black outline delimits the current distribution of *I. alpestris* (IUCN 2021). $T_b$, body temperature.

# 3 Climate warming reshapes Jensen bias (males)

Climate warming altered the magnitude of Jensen bias for all twelve male performance metrics (**Fig. S3.4**), mirroring the pattern observed for females (**Fig. 3**). However, as for females, warming did not produce a uniform response: the strongest relative amplification occurred for minimum body temperature (rel. $\Delta|J| = +510\%$; 60% of cells) and maximum body temperature (rel. $\Delta|J| = +215\%$; 62% of cells), whereas lifespan ($-1\%$) was attenuated. The sign of $J$ changed in 5–46% of cells under $+4\,^{\circ}\mathrm{C}$ warming, with highest rates for basking time (46%), foraging time (43%) and minimum reserve (31%). Even under $+2\,^{\circ}\mathrm{C}$ warming, the direction of change was already discernible for most metrics: 11 metrics showed increased $|J|$, with relative changes ranging from +10% (maximum body length) to +126% (minimum body temperature), while 1 metric was attenuated (minimum reserve ($-1\%$)), and 1 metric (lifespan) switched from amplification at $+2\,^{\circ}\mathrm{C}$ to attenuation at $+4\,^{\circ}\mathrm{C}$, suggesting a nonlinear threshold in the averaging bias. For comparison, in females 1 metric was attenuated under $+2\,^{\circ}\mathrm{C}$ (minimum reserve ($-5\%$)).

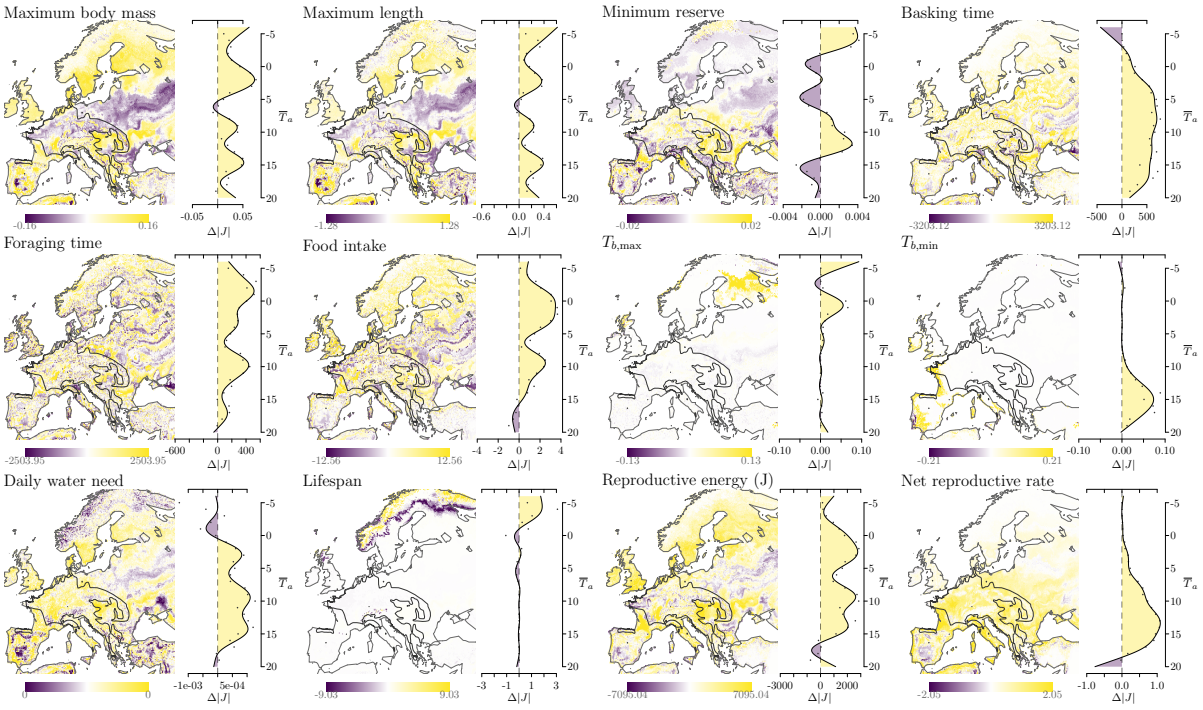



**Figure S3.4:** Jensen bias amplification under $+4\,^{\circ}\mathrm{C}$ warming for males. Each map displays the change in absolute Jensen bias ($\Delta|J|$, left, with colour bar) for each performance metric. The curve (descriptive GAM smoother) at the right of each map represents how $\Delta|J|$ changes along the mean annual temperature ($\overline{T}_a$) gradient. *Yellow* shading (positive $\Delta|J|$) indicates amplification of the averaging bias under warming, while *purple* shading (negative $\Delta|J|$) indicates attenuation. The black outline delimits the current distribution of *I. alpestris* (IUCN 2021). $T_b$, body temperature.

# 4 Consequences for ecological forecasts (males)

Jensen bias produced trait-specific differences between population-average and individual-level forecasts for males under both current and future climates (**Fig. S3.5**), mirroring the female pattern (**Fig. 4**). Reproductive energy exhibited the largest discrepancies, with the population-average model generally overpredicting relative to the individual-level mean under the current climate. The population-average model also overpredicted lifespan (84% of cells; median relative bias +2.1%) and maximum body mass (95% of cells; median +5.5%), and the median relative bias in reproductive energy under +4 °C was +10.9%. Lifespan showed a consistent tendency for the population-average model to predict higher values than the individual-level mean across climate scenarios, whereas maximum body mass showed comparatively close agreement, indicating relatively weak consequences of physiological averaging for this trait. The relative bias maps (**Figs. S3.6, S3.7**) express the same comparison as a percentage of the individual-level mean, making the spatial structure of the disagreement visible.

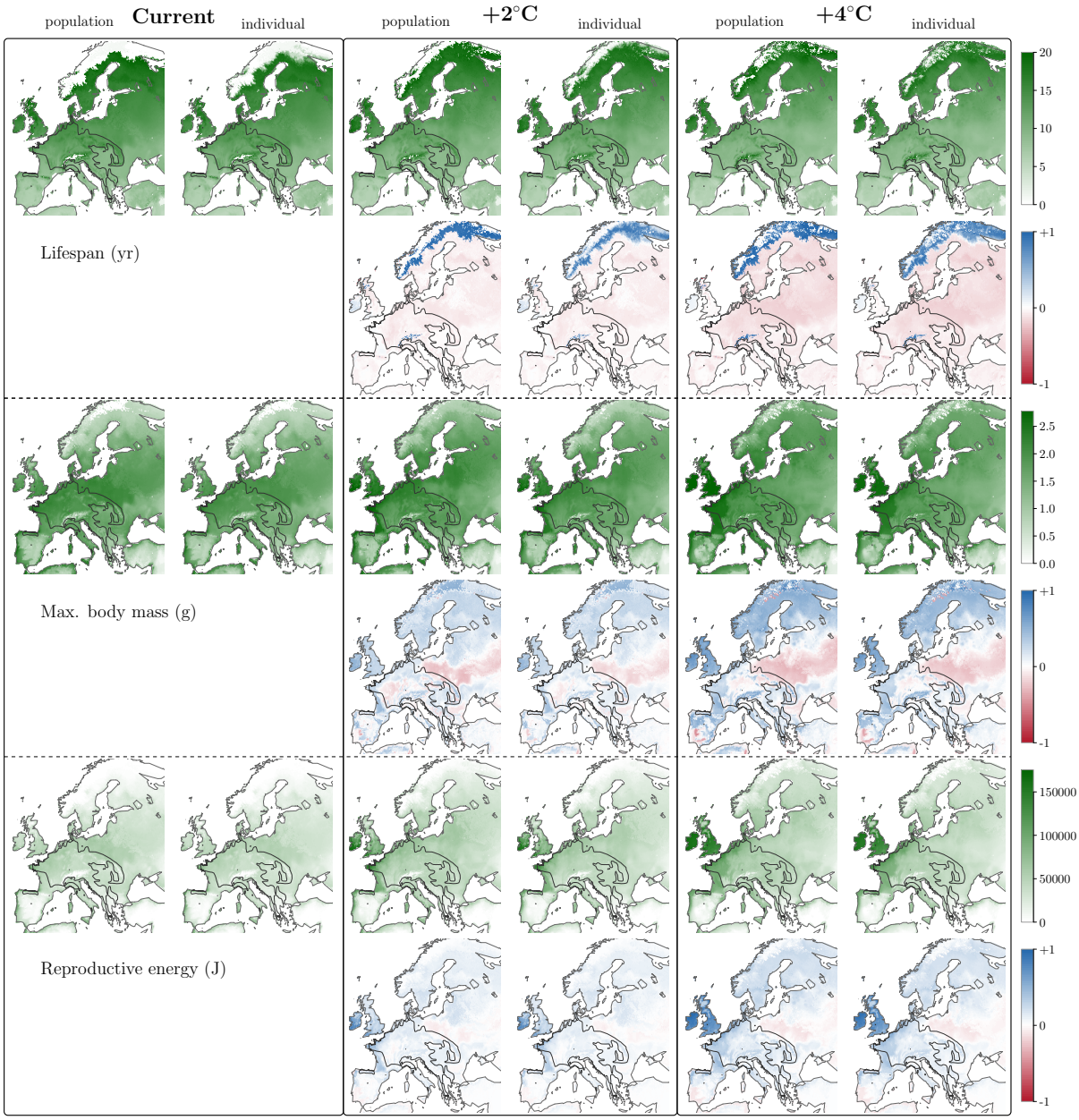



**Figure S3.5:** Comparison of population-average and mean individual-level predictions for lifespan, maximum body mass and reproductive energy for males under current, +2 °C and +4 °C climate scenarios. For each metric, the top row shows the raw predicted values for all six parameterisation–climate combinations on a shared scale. At the bottom, we show the change relative to current climates for the warming scenarios ($P_{\text{future}} - P_{\text{current}}$) on a normalised scale $[-1, 1]$ (*blue* indicates decrease, while *red* indicates increase). The black outline delimits the current distribution of *I. alpestris* (IUCN 2021).

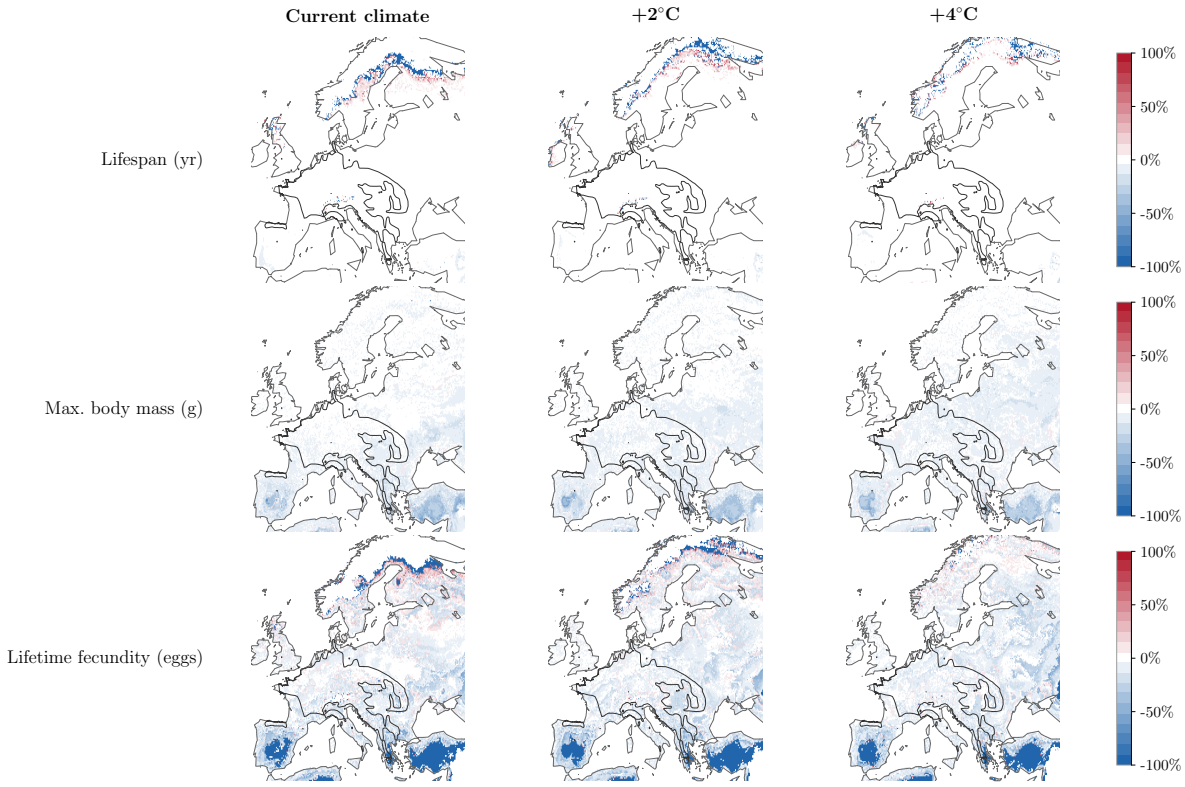



**Figure S3.6:** Relative bias (%) between population-average and individual-level predictions for females across current, +2 °C and +4 °C scenarios. Relative bias is computed as $R = (P_{\text{pop}} - \overline{P}_{\text{ind}})/|\overline{P}_{\text{ind}}| \times 100$ for each grid cell. Blue indicates underestimation by the population-average model ($R < 0$); red indicates overestimation ($R > 0$); white indicates close agreement ($R \approx 0$).

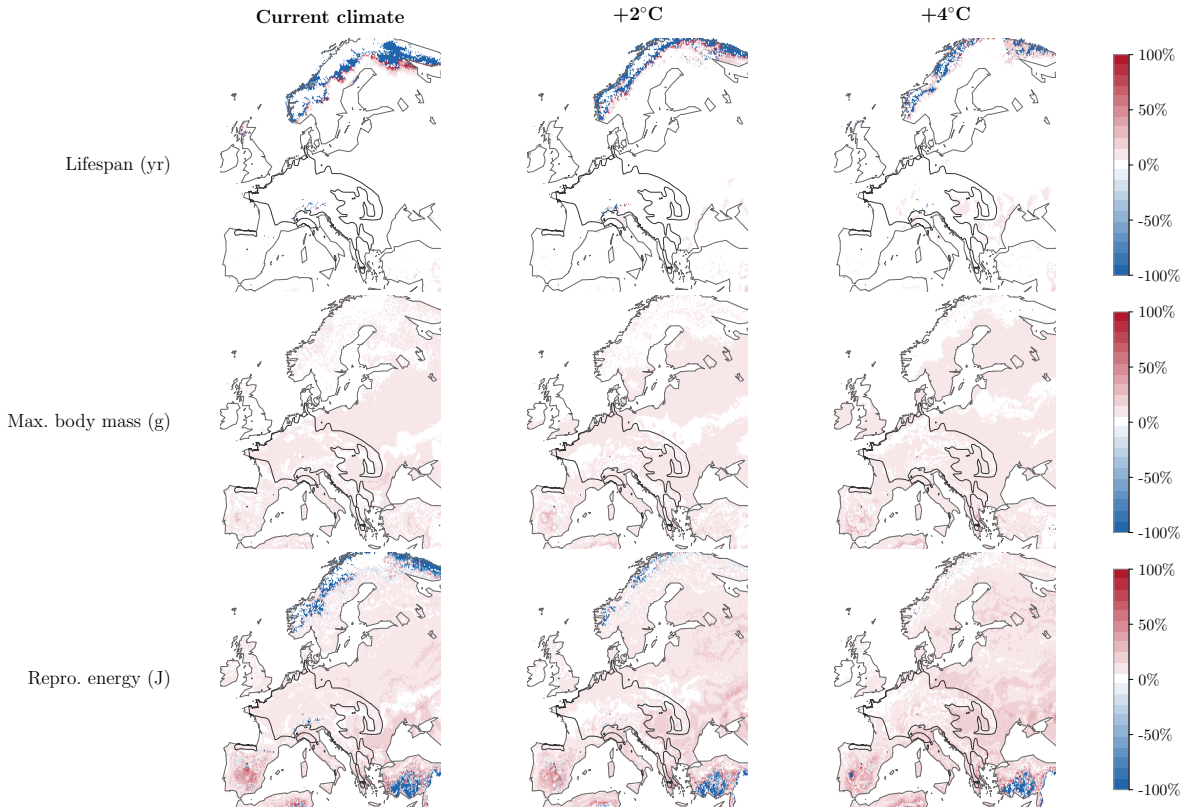



**Figure S3.7:** Relative bias (%) between population-average and individual-level predictions for males, as in **Fig. S3.6** but showing lifespan, maximum body mass and reproductive energy.

# 5 References

IUCN (2021). *Ichthyosaura Alpestris: IUCN SSC Amphibian Specialist Group: The IUCN Red List of Threatened Species 2022: E.T59472A89702213*. International Union for Conservation of Nature.